

2023 Bi H2 Q14

Section: Sustainability and Interdependence

Topic: Symbiosis

Question Summary

Dodder is a parasitic plant that wraps around blueberry bushes and removes sugar and other nutrients from their stems. You are asked to identify and justify the type of symbiosis, then use yield data to calculate the reduction in crop yield and explain control and design features.

Worked Solution

(a)(i) Identify the type of symbiosis

■ Parasitism.

(a)(ii) Justify your answer

Dodder benefits by taking sugars and nutrients from the blueberry plant, while the blueberry is harmed with reduced resources and yield. Benefit to one partner and harm to the other defines parasitism.

(b)(i) Calculate the reduction in blueberry yield due to dodder infection

Average yields: Infected = 18,000 kg per hectare; Uninfected = 22,500 kg per hectare.

Difference per hectare = $22,500 - 18,000 = 4,500$ kg per hectare.

Each field area = 6.7 hectares.

Total reduction = $4,500 \times 6.7 = 30,150$ kg.

■ Answer (b)(i): 30,150 kg

(b)(ii) Explain why the uninfected field is included as a control

To provide a baseline for comparison so that any difference in yield can be attributed to dodder infection rather than other factors. The control keeps all other conditions comparable while removing the infection.

(b)(iii) Why randomising many small plots is good experimental design

Randomising treatments across many small plots increases reliability and reduces bias by averaging out spatial and environmental differences. It increases replication and helps ensure that observed effects are due to the treatment and not to site variation.

Revision Tips

- Parasitism: one organism benefits, the other is harmed.
- Mutualism: both benefit. Commensalism: one benefits, the other is unaffected.

- Always include a control to isolate the effect of the independent variable.
- Good design uses replication, randomisation, and control of confounding variables.

Dodder

Tendrils

Haustorium

Blueberry stem

